

DATABASE OBJECTS

DBMS EXAM & PLACEMENT REVISION NOTES

GATE / INTERVIEW READY

1. WHAT ARE DATABASE OBJECTS?

A **database object** is a named data structure component used to store, represent, generate, access, or manage data within a database management system.

 **Super Memory Trick: T V S I S** $\Rightarrow$ **Table** $\rightarrow$ **View** $\rightarrow$ **Sequence** $\rightarrow$ **Index** $\rightarrow$ **Synonym**

2. OBJECT #1: TABLE (RELATION)

A **Table** is the primary physical database object used to store data structured in rows and columns.

Example: Student Table

Roll No	Name	Marks
101	Husen	90
102	Ali	85
103	John	88

Key Terminology ★

- **Attribute (Column):** Vertical entity representing a property. (e.g., *Roll No*, *Name*)
- **Tuple (Row):** Horizontal entry representing a single complete record. (e.g., *101 | Husen | 90*)

Degree = Total Columns

The count of attributes/columns in a relation.

Example: *Student(RollNo, Name, Marks, Age)* $\Rightarrow$ **Degree = 4**

Cardinality = Total Rows

The count of tuples/rows in a relation.

Example: A table with 100 student records $\Rightarrow$ **Cardinality = 100**

3. OBJECT #2: VIEW (VIRTUAL TABLE)

A **View** is a logical/virtual representation of data derived from one or more underlying base tables. It does *not* store data physically.

```
CREATE VIEW Student_View AS
SELECT RollNo, Name FROM Student;
```

Why Use Views?

- **Restrict Data Access:** Hide sensitive columns.
- **Simplify Queries:** Encapsulate complex JOINS.
- **Data Independence:** Present customized perspectives of data.

4. OBJECT #3: SEQUENCE (NUMBER GENERATOR)

Definition & Purpose

A **Sequence** automatically generates sequential integer values (1, 2, 3, 4...).

Real-World Use: Exam registration IDs, transaction numbers, receipt acknowledgements.

Primary Key Integration

Primary keys require **Unique + NOT NULL** values. Sequences automatically generate unique sequential keys for new insertions without manual intervention.

Sequence $\rightarrow$ 101 $\rightarrow$ 102 $\rightarrow$ 103

5. OBJECT #4: INDEX (QUERY ACCELERATOR)

An **Index** is a performance-tuning structure used to speed up data retrieval operations (specifically DML SELECT queries) by minimizing secondary storage I/O block accesses.

✗ Without Index

Database performs full table scans → Searches many disk blocks → High I/O cost → **Slow Retrieval**.

⚡ With Index

Database queries index first → Pinpoints exact data block location → Low I/O cost → **Fast Retrieval**.

6. TYPES OF DATABASE INDEXES

Primary Index

Created on an **Ordered Data File** using a **Key Field** (Primary Key).
Unique search key values.

Clustered Index

Created on an **Ordered Data File** using a **Non-Key Field**. Duplicate values grouped together.

Secondary Index

Created on an **Unordered Data File** for either Key or Non-Key search attributes.

Dense vs. Sparse Index

Structure

- **Dense Index:** Contains an index entry for *every single record* in the main data file. (More space, faster search).
- **Sparse Index:** Contains index entries for *data blocks only* rather than individual records. (Smaller size).

B-Tree & B+ Tree Indexing

Tree Structures

Self-balancing search tree structures that maintain sorted data for logarithmic lookup time ($O(\log N)$).

- **B-Tree:** Data pointers stored in internal + leaf nodes.
- **B+ Tree:** Data pointers stored *only* in doubly-linked leaf nodes.

7. OBJECT #5: SYNONYM (OBJECT ALIAS)

Definition & Purpose

A **Synonym** creates an alternative, permanent alias name for any database object.

Primary Purpose: Simplifies references to complex or lengthy object names and provides abstraction across schemas.

Example Concept

Very_Long_Student_Table_Name

↓ (**Synonym**)

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What can a Synonym point to? Tables, Views, Sequences, Stored Procedures, Packages, or other Synonyms.

 8. ALL 5 DATABASE OBJECTS AT A GLANCE

Object	Icon	Primary Function	Key Characteristics	Memory Keyword
Table		Stores physical data	Organized into rows (tuples) and columns (attributes)	Data Store
View		Virtual data representation	Logical view, contains no physical data of its own	Logical Mask
Sequence		Generates sequential numbers	Used to automate Primary Key / ID creation	Auto Number
Index		Accelerates query execution	Reduces disk I/O costs during data retrieval	Query Speed
Synonym		Provides an object alias	Simplifies long names & masks object location	Alias Name

 9. PLACEMENT & GATE INTERVIEW Q&A

Q1. What are the 5 core database objects discussed?

Table, View, Sequence, Index, and Synonym.

Q2. Differentiate clearly between Degree and Cardinality.

Degree is the total number of **attributes (columns)** in a table, whereas Cardinality is the total number of **tuples (rows)** in that table.

Q3. Does a View occupy physical storage space for table data?

No. A view only stores its query definition in the data dictionary. The data itself is dynamically generated from underlying base tables upon query execution.

Q4. How does an Index improve DML SELECT query performance?

It creates an organized lookup structure that points directly to the physical disk address of records, bypassing full-table sequential scans and lowering I/O cost.

Q5. Differentiate between a Dense Index and a Sparse Index.

A Dense Index has a search key entry for every record in the data file, whereas a Sparse Index has entries only for block anchors/data blocks.

 10. 30-SECOND RAPID REVISION SUMMARY

Architectural Map

DATABASE OBJECTS

- |— TABLE → Stores Data (Rows/Cols)
- |— VIEW → Represents Virtual Data
- |— SEQUENCE → Auto-Generates Unique IDs
- |— INDEX → Speeds Up Lookups
- |— SYNONYM → Alternative Alias

 One-Line Master Rule

"TABLE stores data,
VIEW represents data,
SEQUENCE generates IDs,
INDEX speeds up queries, and
SYNONYM renames objects."

